

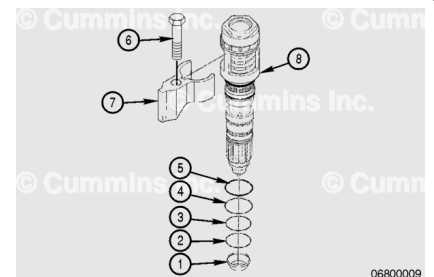
Trainings ▾

General Information

with Mechanically Actuated Injector

The injector consists of the following components:

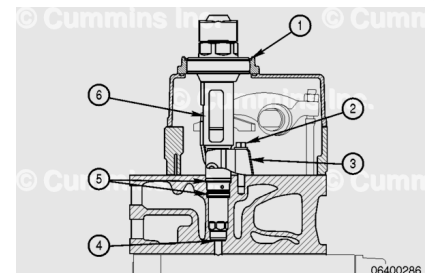
1. Sealing ring
2. O-ring (brown, small)
3. O-ring (green)
4. O-ring (light blue)
5. O-ring (black, large)
6. Capscrew
7. Injector hold-down clamp
8. Injector.



with Electronically Actuated Injector

The injector consists of the following components:

1. O-ring and damper
2. Capscrew
3. Injector hold-down clamp
4. Seal
5. O-rings
6. Injector.



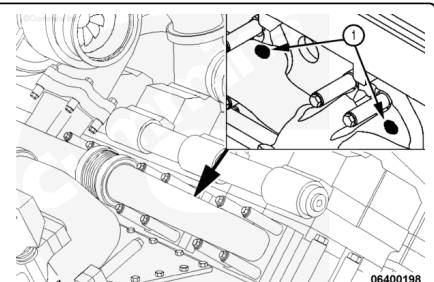
Initial Check

with Mechanically Actuated Injector

Misfire Using Heat Indicator Marker

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Note : This procedure is **not** effective when the symptoms occur **only** at low idle or when the engine is cold.

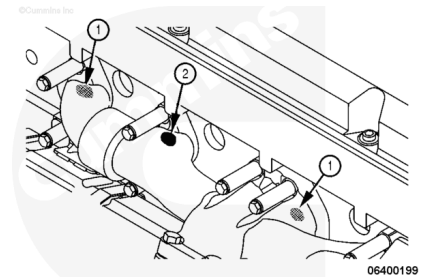
If the engine exhibits symptoms of an injector **not** firing, perform the following check:

- Operate the engine to normal operating coolant temperature of 80°C [180°F].
- Stop the engine and allow the exhaust manifold to cool for 10 minutes.
- Use a 204°C [400°F] temperature indicator marker, Part Number 3165163, or equivalent, to apply a mark (1) on the surface of all exhaust manifold ports. The mark **must** be as close to the cylinder head as possible, but **not** directly on the exhaust manifold mounting flange.
- Start the engine and operate under light load for a short time. Do **not** operate the engine under heavy load or for an extended period of time, as this can cause inaccurate results.

Note : It is not recommended to use the temperature indicator markers to troubleshoot for cylinders that are operating too hot.

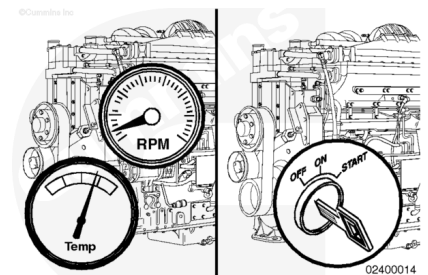
If the 204°C [400°F] mark melts (1) the color will disappear and leave **only** a transparent glazed appearance where the mark was. This indicates that the cylinder is operating at a temperature above 204°C [400°F].

If the 204°C [400°F] mark does **not** melt (2), the mark color remains present. This indicates that the cylinder is **not** operating at a temperature of 204°C [400°F], indicating a possible injector malfunction.



⚠ CAUTION ⚠

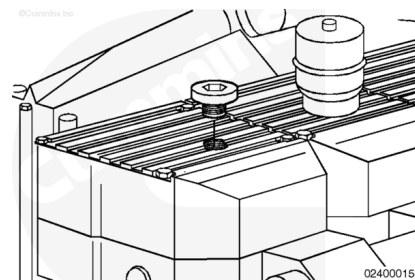
This tool is not intended to be used above the engine idle speed. Do not use above 900 rpm engine speed. Doing so can damage the engine. Start and operate the engine until the coolant temperature reaches the normal range of 80°C [180°F]. Shut the engine off.



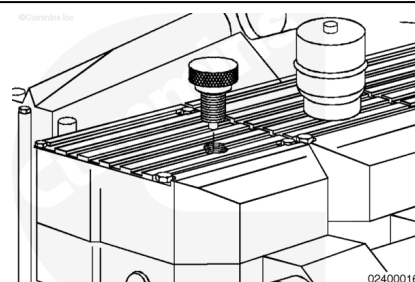
Start and operate the engine until the coolant temperature reaches the normal range of 80°C [180°F].

Shut the engine OFF.

Use a 12-mm hex socket or Allen wrench to remove the existing plug from the top side of the cast iron valve cover.

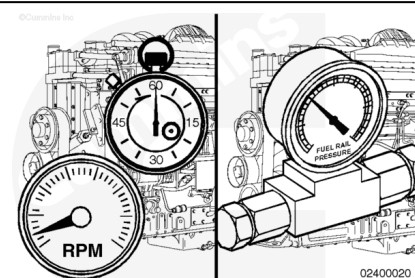


Screw the single injector diagnostic cut-out tool **clockwise** into the threaded hole in the valve cover until it bottoms out on the valve cover.



⚠ CAUTION ⚠

This tool is not intended to be used above the engine idle speed. Do not use above 900 rpm engine speed. Doing so can damage the engine.



Install a 0 to 207 kPa [0 to 30 psi] pressure gauge on the Compuchek™ fitting located at the front of the block.

Start and operate engine at idle speed for approximately 1 minute.

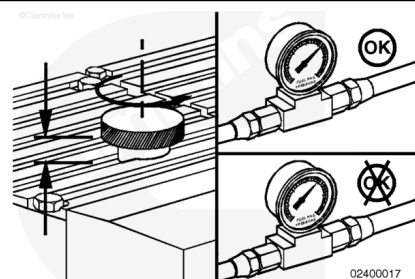
- Record the fuel rail pressure reading.

Rotate the single-injector diagnostic cut-out tool **counterclockwise** until it is approximately 13 mm [½ inch] clear of the valve cover.

- Note the fuel rail pressure reading.

If the cylinder being tested is working properly, the fuel rail pressure can decrease on reading number 2.

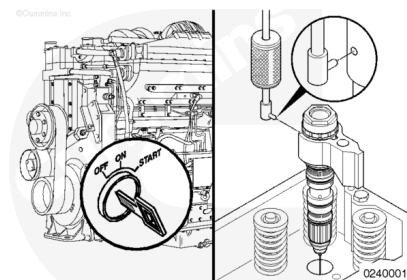
If the cylinder being tested is weak or **not** firing, the fuel rail pressure will **not** change between reading number 1 and number 2.



Shut the engine OFF.

Repeat the above steps in different rocker lever covers until the faulty injector is found.

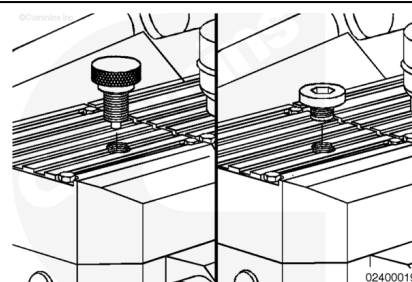
Replace the malfunctioning injector(s).



After engine diagnostic testing, remove the single-injector diagnostic cut-out tool from the valve cover and install the plug.

Tighten the plug.

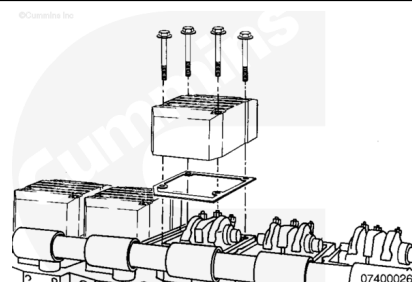
Torque Value: 45 n•m [33 ft-lb]



Check the injectors when it is suspected they were run dry (without fuel).

Remove the rocker lever covers and check to be sure all of the injector links are intact. If they are intact, install the covers and follow the normal injector inspection procedure. If they are **not** intact, follow the injector inspection step in this procedure.

Warm up the engine by running at high idle to stabilize the exhaust temperature. If the exhaust temperatures are within 38°C or 68°F, all injectors are firing and working correctly.

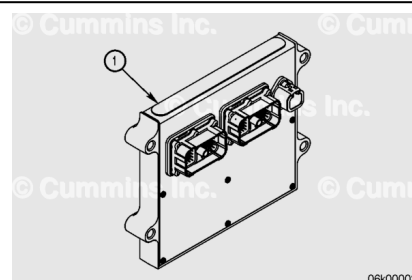


with Electronically Actuated Injector

Misfire Using Injector Tester Tool

Note : This test **must** be performed with the engine shut down for at least 5 minutes.

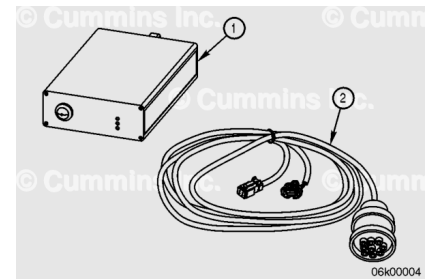
Note : If the engine is equipped with CM2150 engine control modules (ECMs) (1), cover the plug with a lint-free cloth. Slowly loosen the plug on the last injector of the high-pressure fuel lines 1/4 to 1/2 turn. The plug does **not** need to be removed to relieve the pressure. Connect INSITE™ electronic service tool and verify the fuel pressure has bled down by monitoring the fuel rail pressure.



If the engine exhibits symptoms of an injector **not** firing, perform the following check using the injector tester (1)(2), Part Number 2892293:

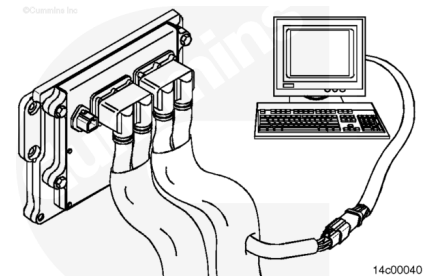
1. Plug the tool harness into the injector tester tool via the 4-pin connector.
 2. Plug the tool harness into the engine harness via the 9-pin connector.
 3. Verify injector tester tool yellow READY light is illuminated. If **not**, verify the power supply to the 9-pin connector. If illuminated, proceed.
 4. Turn keyswitch ON.
 5. Plug the tool harness into the injector via the 2-pin connector.
 6. Fully depress and release the injector tester tool PRESS TO TEST button.
- If a green PASS light is illuminated, the injector is functioning.
 - If a red FAIL light is illuminated, the injector is failed and **must** be replaced.

Repeat steps 5 and 6 until all injectors have been tested.



Misfire Using Injector Cutout Tool

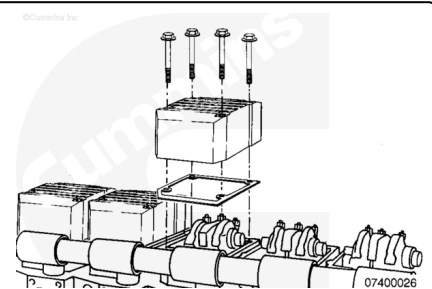
Use INSITE™ electronic service tool to perform the Cylinder Cutout Test on engines with electronically actuated injectors. See INSITE™ electronic service tool manual for instructions.



Check the injectors when it is suspected they were run dry (without fuel).

Remove the rocker lever covers and check to be sure all of the injector links are intact. If they are intact, install the covers and follow the normal injector inspection procedure. If they are **not** intact, follow the injector inspection step in this procedure.

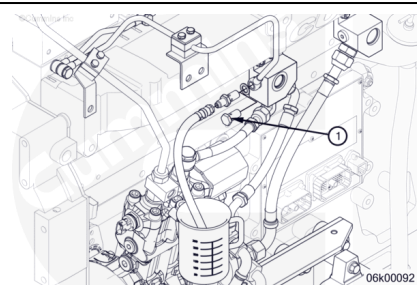
Warm up the engine by running at high idle to stabilize the exhaust temperature. If the exhaust temperatures are within 38°C or 68°F, then all injectors are firing and working correctly.



High-Pressure Injector Return Flow Test with Electronically Actuated Injector

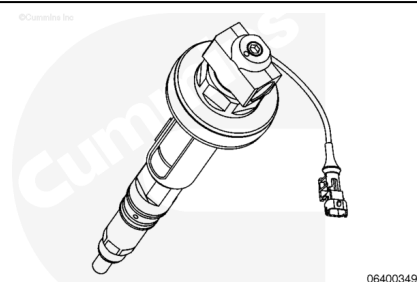
Remove the banjo bolt (1) from the fuel injector drain line at the fuel return manifold. Install the fuel drain fitting and hose, Part Numbers 3164614 and 2892248, in place of the banjo bolt.

Place the end of the fuel injector drain line fuel return flow hose in a graduated cylinder with at least 100 ml [3.4 oz] capacity to collect return flow.



Unplug all six of the injector connectors and crank the engine at least 150 rpm for 30 seconds.

Note : Do not crank the engine for more than 30 seconds; otherwise damage to the starter may occur. Wait 5 minutes before the test is repeated.



Flow Specification: Engine Cranking

19L (Six Injectors)

75 ml per 30 seconds [2.5 fl-oz per 30 seconds]
maximum

If fuel flow is greater than specification, it is necessary to perform the Injector Isolation Test described below.

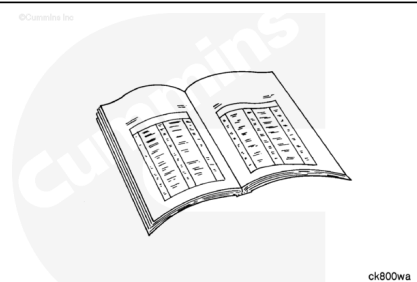


High-Pressure Injector Return Flow Isolation Test with Electronically Actuated Injector

Remove the fuel drain manifold banjo bolts and remove both pieces of the fuel drain manifold. Refer to Procedure 006-013 in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-013-tr.html>)

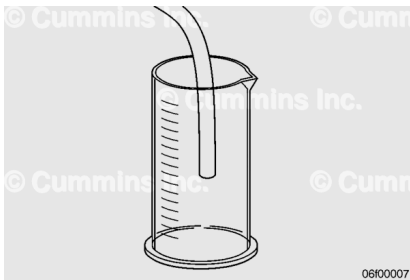
Note : Marine engines will have to remove the intake manifold to gain access to the drain manifold. Refer to Procedure 010-023 in Section 10.

(</qs3/pubsys2/xml/en/procedures/20/20-010-023-tr.html>) Refer to Procedure 010-019 in Section 10.



Install a fuel drain test fitting and sealing washer, Part Numbers 2892297 and 3963990, into each of the cylinder heads.

Slide a hose over the nipple on each of the fuel drain test fittings and route it to a graduated cylinder with at least 100 ml [3.4 oz] capacity.

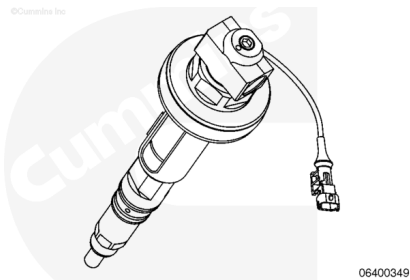


Install threaded plug, Part Number 3678610, into the fuel drain block.

Unplug all six of the injector connectors and crank the engine at least 150 rpm for 30 seconds.

Note : Do not crank the engine for more than 30 seconds; otherwise damage to the starter may occur. Wait 5 minutes before the test is repeated.

Record the amount of fuel collected from each of the cylinders.

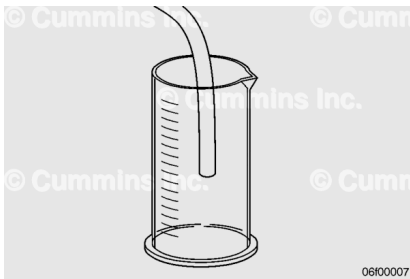


Flow Specification: Engine Cranking

19L (One Injector)	15 ml per 30 seconds [0.5 fl-oz per 30 seconds] maximum
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Replace the injector on any cylinder that has more leakage than specified.

If any injectors were replaced, repeat the test on all of the six cylinders until all cylinders have less flow than specified.



Remove the fuel drain hoses and test fittings and reassemble the fuel drain lines. Refer to Procedure 006-013 in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-013-tr.html)



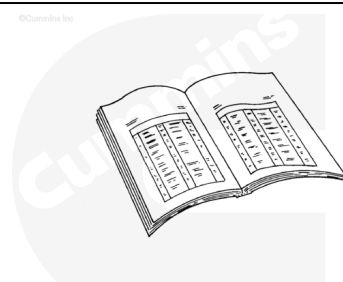
Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

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ck800wa

⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.

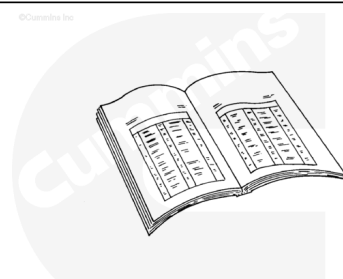
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html)
- Remove the rocker lever assembly. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html)

with Electronically Actuated Injector

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

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ck800wa

⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.

- Remove the injector supply lines. Refer to Procedure 006-051 in Section 6.
(/qs3/pubsys2/xml/en/procedures/20/20-006-051-tr.html)
- Install the injector protective plugs, Part Number 4918767.
- Remove the two-piece valve cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html)

Remove

with Mechanically Actuated Injector

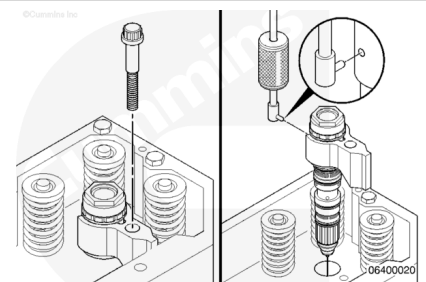
Note : The hold-down clamp can **not** be removed until the injector is removed.

Remove the injector hold-down capscrew. Remove the injector and clamp.

Use injector puller, Part Number 3823579, or equivalent, to remove the injectors. Insert the pin of the tool in the hole provided in the body of the injector. The hole faces the front of the engine.

Note : If the injector puller is **not** available, carefully use a pry bar. Pry upward on the injector against the cylinder head.

Cover the injection bore to prevent foreign objects from entering the cylinder.

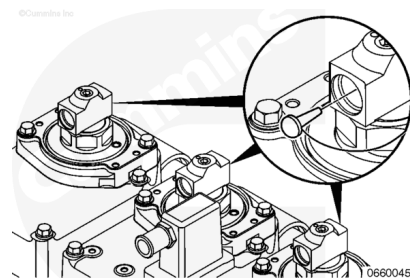


with Electronically Actuated Injector

Note : Do not reuse protective plugs. The protective plugs must be used immediately upon removal from the plastic wrapping. If they are **not** used immediately, the protective plugs must be discarded. If the protective plugs are fouled in any manner before use,

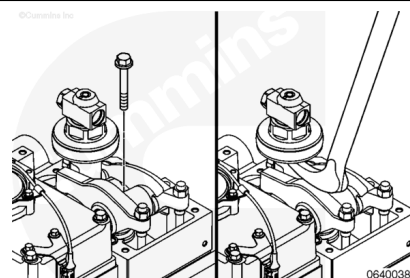
the protective plugs **must** be discarded and a new set used.

If **not** already completed, plug the ports in each side of the injector T-piece with the injector protective plugs, before removing the injector. Make sure that the plugs are wedged firmly into place. This will prevent debris from entering and causing damage to the injector.



⚠ CAUTION ⚠

Do not reuse the injector hold-down capscrew or injector combustion seal. The reuse of those components can result in engine damage.



Note : On applications with high drain line head pressure, such as a fuel tank mounted higher than the engine, shut off the fuel drain line valve and evacuate the fuel drain line of fuel before removing the injector.

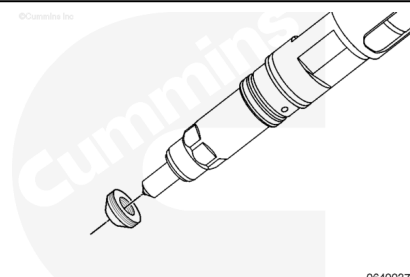
Remove and discard the injector hold-down capscrew.

Use a heel bar to remove the injector. Place the heel between the injector body and the exhaust rocker lever and gently pry the injector from the injector bore.

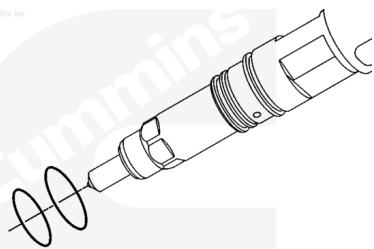
Note : On applications with high drain line head pressure, such as a fuel tank mounted higher than the engine, it is possible for fuel to leak from the drain line into the cylinder when the injector is removed. After removing the injector, check for fuel dripping into the cylinder and evacuate the fuel before continuing in the removal procedure.

Cover the injector bore in the cylinder head to prevent foreign objects from entering and causing damage to the bore and combustion chamber.

Remove and discard the seal on the end of the injector.



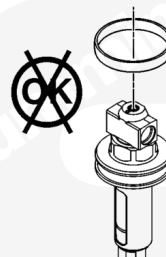
Remove and discard the two o-rings on the injector.



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Do **not** remove the o-ring/damper at this time.

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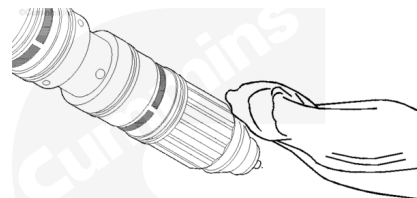


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Clean and Inspect for Reuse

with Mechanically Actuated Injector

Use a lint-free cloth to clean the exterior of the injector.



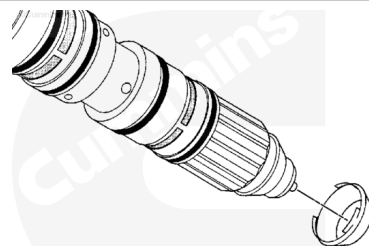
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Remove the sealing ring. Note cylinder location and thickness.

Discard sealing ring

Note : Oversize sealing rings are available to adjust injector protrusion in resurfaced and Recon® cylinder heads. Refer to Procedure 002-022 in Section 2. (</qs3/pubsys2/xml/en/procedures/20/20-002-022.html>)

Note : Recon® cylinder heads are marked to identify which sealing ring should be used. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/20/20-002-004-tr.html>)



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The o-rings **must** be installed in the correct order on the injector. Take note of the o-ring color and orientation prior to removal.

Remove o-rings 1, 2, 3, and 4.

Use a lint-free cloth. Clean the exterior of the injector.

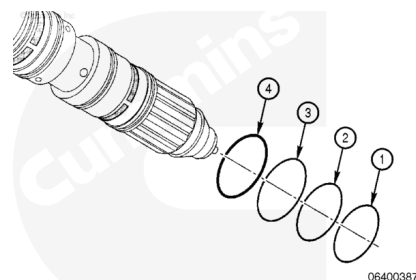
Carefully check the area where the sealing ring touches the injector.

Inspect o-ring grooves for damage.

Inspect injector body and cup retainer for cracks or other damage.

Inspect injector links for excessive wear, pitting or scoring on the ball ends, and other damage.

Note : If the link is damaged or pitting and scoring can be seen or felt, the link **must** be replaced.

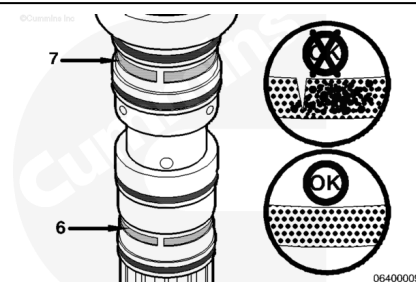


⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

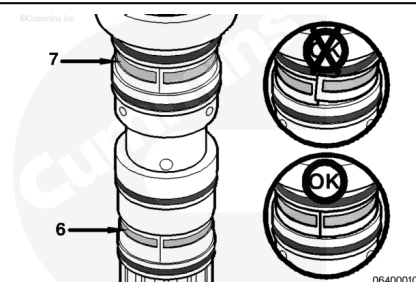
Use solvent when cleaning the filters. The screens **must** be thoroughly dry prior to installation.

Check the rail (6) and timing (7) filter screens for debris, tears, or punctures. Replace the filter screens if damaged or dirty.

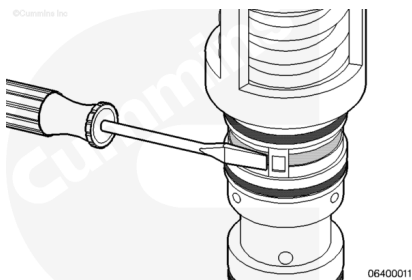


Check the rail (6) and timing (7) filter screens for proper installation.

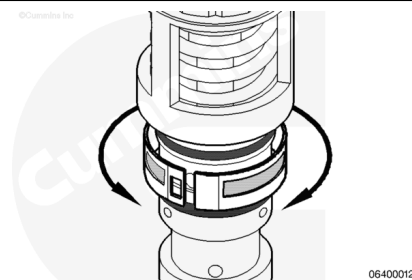
The filter screens **must** be securely snapped into the correct position.



Use a small screwdriver to remove the filter screen by gently prying up on the connector clip.

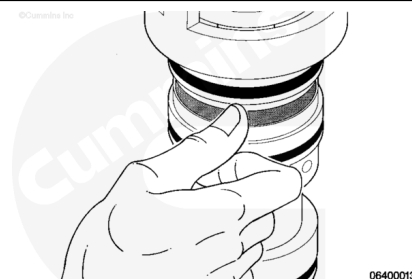


Wrap the filter screen around the inlet groove.



A properly secured connector will create a single snapping sound.

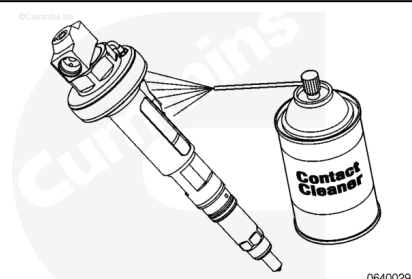
Apply pressure on the filter screen connector clip.



with Electronically Actuated Injector

⚠ WARNING ⚠

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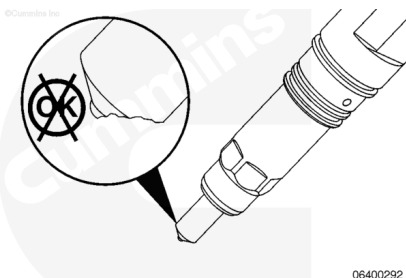


Clean the injector assembly with electrical contact cleaner, Part Number 3824510, or equivalent. Allow to air dry completely.

Do **not** use compressed air to dry or clean the injector.

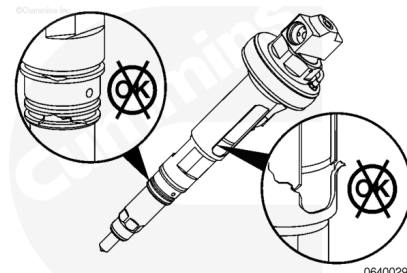
Inspect the nozzle tip for damage.

Replace injectors with damaged nozzles that prevent fuel delivery into the combustion chamber.



Inspect the injector body for signs of corrosion, cracks, and other damage.

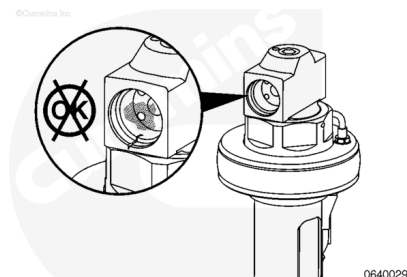
Inspect the o-ring grooves for damage that will **not** allow the o-rings to seal properly.



Inspect the T-fitting on top of the injector where the injector supply lines attach.

Inspect for corrosion, cracks, and other damage that does **not** allow proper sealing of the injector supply lines.

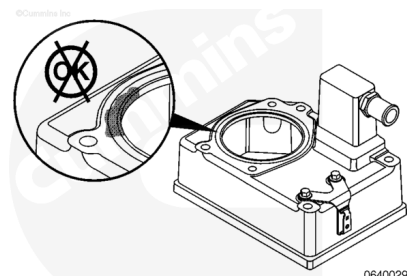
Replace damaged injectors.



Inspect the o-ring/damper for wear, cracks, and other damage. Replace if damaged.

If the valve cover shows signs of oil leakage around the injector, replace the o-ring/damper.

Inspect the injector wiring for damage to the wires, shrink tubing, elbow, and connector.



Install

with Mechanically Actuated Injector

Do **not** lubricate the o-rings until the injector is ready for installation in the cylinder head.

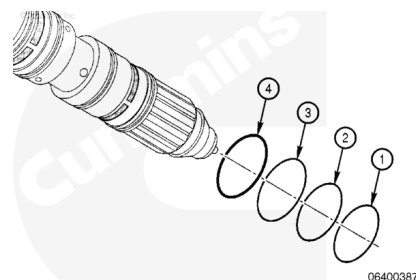
Install four new o-rings on the injector. Do **not** twist the o-rings.

O-ring (1) is brown and is smaller in diameter than the other three.

O-ring (2) is green.

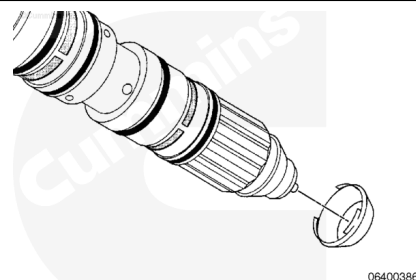
O-ring (3) is light blue and is larger in diameter than (1), but has a smaller cross section than o-ring (4).

O-ring (4) is black and is the largest in diameter and cross section.

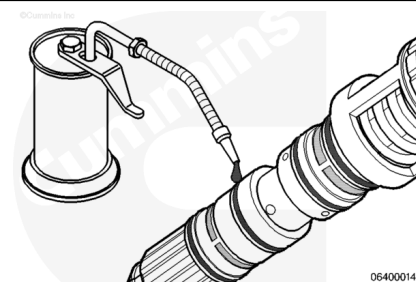


Install a new, proper thickness sealing ring which corresponds to the cylinder location noted during removal.

Note : If installing an injector for the first time into a resurfaced cylinder head, use the following procedure to determine proper sealing ring thickness. Refer to Procedure 002-022 in Section 2. (</qs3/pubsys2/xml/en/procedures/20/20-002-022.html>)



Use clean lubricating engine oil (15W-40) to lubricate the o-rings.



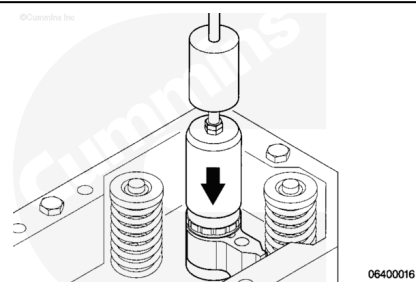
Install the hold-down clamp on the side of the injector.

Install the injector and hold-down clamp in the cylinder head.

Note : If the injector installation tool, Part Number 3824830, is not available, use a 40 mm [1 9/16 in] socket with an extension and a rubber mallet to install the injector.

Use injector installation tool, Part Number 3824830, to set the injector in the bore.

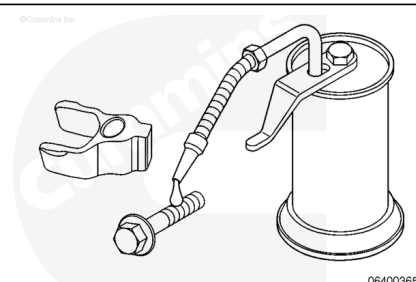
The slide hammer will make a dull sound when the injector is seated properly.



Use clean engine oil to lubricate the injector hold-down capscrew.

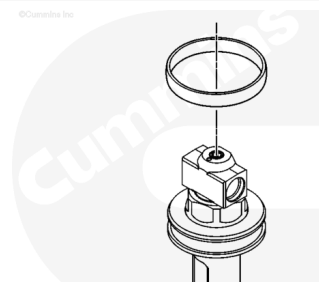
Install and tighten the injector hold-down capscrew.

Torque Value: 54 n•m [40 ft-lb]



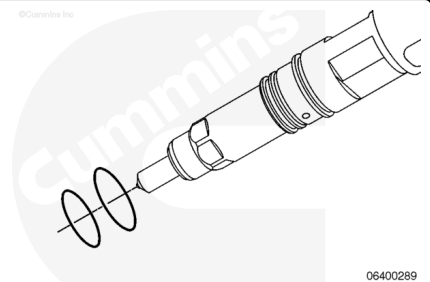
with Electronically Actuated Injector

Install the new o-ring/damper on the injector.



06800270

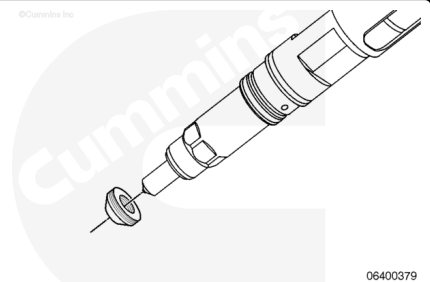
Install the new o-rings on the injector body.



06400289

⚠ CAUTION ⚠

Do not reuse the injector hold-down capscrew or injector combustion seal. The reuse of those components can result in engine damage.

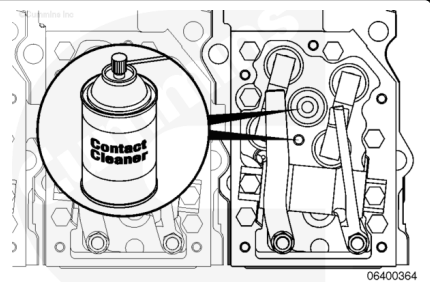


06400379

Install the new seal on the end of the injector.

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.



06400364

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Keep the injector bore and combustion chamber protected when using compressed air to prevent contamination. Failure to do so can cause engine damage.

⚠ CAUTION ⚠

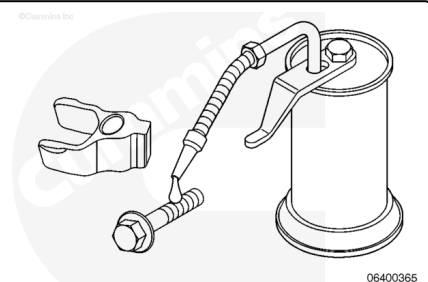
Do not reuse the injector hold-down capscrew or injector combustion seal. The reuse of those components can result in engine damage.

Note : On applications with high drain line head pressure, such as a fuel tank mounted higher than the engine, it is possible for fuel to leak from the drain line into the cylinder when the injector is removed. Prior to installing the injector, check for fuel dripping into the cylinder and evacuate the fuel before installing the injector.

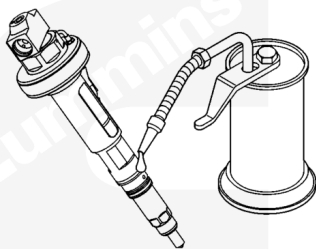
Clean the injector bore in the cylinder head with electrical contact cleaner, Part Number 3824510, or equivalent, to make sure the o-rings will seal properly.

Clean the threaded hole in the cylinder head where the injector hold-down capscrew will be installed with electrical contact cleaner, Part Number 3824510, or equivalent, and compressed air.

Lubricate the threads and underside of the new injector hold-down capscrew with clean lubricating engine oil.



Lubricate all of the o-rings with a light coating of clean lubricating engine oil.

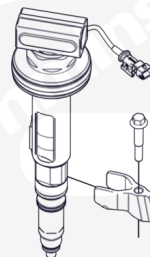


06400366

⚠ CAUTION ⚠

Do not reuse the injector hold-down capscrew or injector combustion seal. The reuse of those components can result in engine damage.

Assemble the injector hold-down clamp and capscrew into the injector.



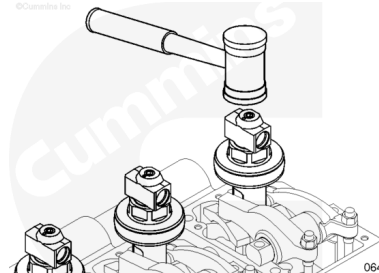
06s00062

⚠ CAUTION ⚠

Only use a soft face mallet. Otherwise damage to the injector can occur.

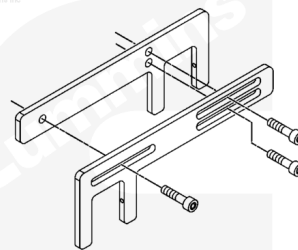
The injector installer can be used on single or multiple injector installations.

Install the assembly into the cylinder head by tapping gently on the injector with a soft face mallet.



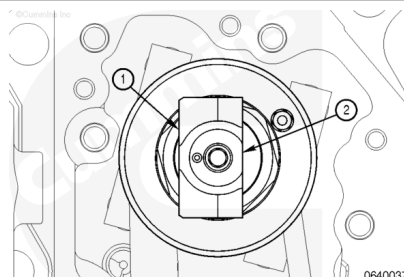
06400368

Assemble the injector installer kit, Part Number 4918620. Hand tighten the three shoulder capscrews.



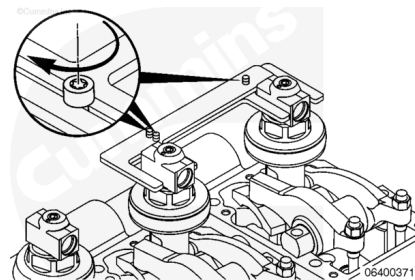
06400369

Place the injector installer on the injector such that the ball spring plunger is on the tapered side of the injector (1). If necessary, adjust the ball spring plunger so that it presses against the injector. Make sure the flat side of the injector (2) sits flush with the injector installer.



06400370

Align the injector installer with two injectors.



Tighten the injector hold-down capscrews.

The 60 degree torque angle can be achieved at the capscrew by marking the location of one corner of the capscrew to a location on the injector hold-down clamp.

Tighten to:

Torque Value: 68 n•m [50 ft-lb]

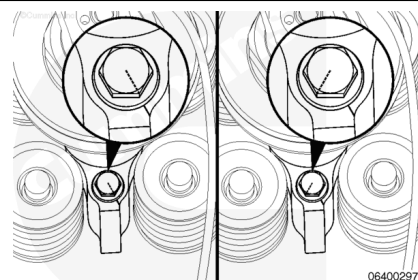
Back off completely.

Tighten to:

Torque Value: 30 n•m [22 ft-lb]

Rotate the capscrew 60 degrees.

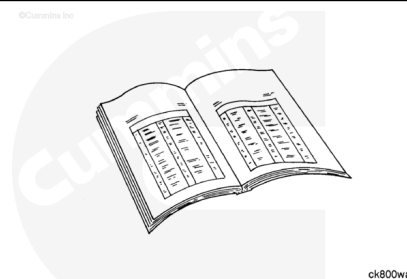
Remove the injector installer.



Finishing Steps

with Mechanically Actuated Injector

- Install the rocker lever assembly. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html)
- Adjust the valves and injectors. Refer to Procedure 003-006 in Section 3.
(/qs3/pubsys2/xml/en/procedures/20/20-003-006-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html)
- Operate the engine and check for leaks.



with Electronically Actuated Injector

- Adjust the valves. Refer to Procedure 003-006 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-006-tr.html>)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html>)
- Remove the protective plugs, Part Number 4918767.
- Install the injector supply lines. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-051-tr.html>)
- Operate the engine and check for leaks.

